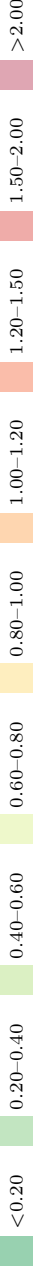


Experiment	Type	Qentaur				Llama-Centaur				Scoltaur				Olmotaur				[binz2025foundation]				Chance
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	L-70Bp	C _r	C _p	Cog _p	ln(k)		
Balloon analog risk task [frey2017risk]	Decision	0.07	0.07	0.07	0.06	0.06	0.07	0.07	0.06	0.08	0.07	0.07	0.07	0.07	0.07	0.08	0.07	0.06	0.09	0.69		
CPC18 [plonsky2018when]	Decision	0.35	0.34	0.34	0.33	0.33	0.36	0.35	0.34	0.42	0.41	0.40	0.34	0.44	0.34	0.41	0.35	0.34	0.66	0.69		
Columbia card task [frey2017risk]	Decision	0.20	0.20	0.20	0.20	0.20	0.21	0.20	0.20	0.20	0.24	0.22	0.21	0.20	0.22	0.20	0.23	0.21	0.26	0.69		
Decisions from description [vulff2018sampling]	Decision	0.62	0.63	0.62	0.61	0.60	0.62	0.63	0.60	0.68	0.66	0.64	0.61	0.63	0.62	0.76	0.59	0.53	0.61	0.69		
Experiential-symbolic task [garcia2023experiential]	Decision	0.48	0.47	0.46	0.46	0.46	0.47	0.47	0.46	0.47	0.45	0.43	0.47	0.47	0.47	0.70	0.46	0.45	—	—		
Gardening task [flesch2018comparing]	Decision	0.39	0.39	0.38	0.38	0.38	0.40	0.39	0.38	0.57	0.54	0.53	0.38	0.58	0.38	0.50	0.49	0.38	0.91	0.69		
Go/no-go [enkavi2019large]	Decision	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00		
Intertemporal choice [ruggieri2022globalizability]	Decision	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.47	0.45	0.44	0.44	0.44	0.44	0.73	0.44	0.43	0.66	0.69		
Multi-attribute DM [hilbig2014generalized]	Decision	0.09	0.08	0.09	0.08	0.08	0.08	0.08	0.08	0.31	0.19	0.16	0.08	0.10	0.09	0.15	0.06	0.06	0.19	0.69		
Risky choice [krueger2024identifying]	Decision	0.48	0.47	0.44	0.43	0.42	0.49	0.46	0.43	0.72	0.61	0.53	0.45	0.58	0.46	0.65	0.43	0.43	—	—		
choices13k [peterson2021using]	Decision	0.44	0.44	0.43	0.43	0.43	0.44	0.44	0.43	0.49	0.46	0.44	0.44	0.45	0.44	0.53	0.43	0.43	0.66	0.69		
Multi-task RL [tomov2021multitask]	MDP	0.60	0.59	0.58	0.58	0.57	0.60	0.59	0.57	0.68	0.66	0.63	0.59	0.78	0.59	0.66	0.57	0.57	1.04	1.10		
Tile-revealing task [kumar2023disentangling]	MDP	0.02	0.01	0.00	0.07	0.00	0.02	0.00	0.00	2.69	1.99	0.04	0.00	0.06	0.09	2.74	2.29	1.87	—	—		
Two-step task [koel2017cost]	MDP	0.54	0.53	0.53	0.53	0.53	0.55	0.53	0.53	0.60	0.60	0.56	0.53	0.57	0.53	0.61	0.53	0.50 [†]	0.60 [†]	0.69		
Two-step task [koel2016when]	MDP	0.51	0.50	0.49	0.49	0.48	0.52	0.50	0.49	0.59	0.58	0.52	0.50	0.55	0.50	0.61	0.48	0.50 [†]	0.60 [†]	0.69		
Two-step task [zotrowitz2023data]	MDP	0.52	0.51	0.51	0.51	0.51	0.52	0.51	0.51	0.60	0.58	0.52	0.52	0.54	0.51	0.61	0.51	0.50 [†]	0.60 [†]	0.69		
Virtual subway network [tomov2020discovery]	MDP	1.16	1.15	1.14	1.13	1.12	1.19	1.16	1.13	1.43	1.33	1.30	1.14	1.32	1.15	1.53	1.16	1.13	—	1.61		
Zoopermarket [ludwig2023human]	MDP	0.54	0.52	0.51	0.50	0.48	0.54	0.52	0.50	0.63	0.61	0.56	0.51	0.61	0.52	0.60	0.50	0.48	0.60	0.69		
Cond. assoc. learning [collins2014working]	Memory	0.54	0.53	0.52	0.52	0.51	0.55	0.53	0.52	0.74	0.64	0.58	0.53	0.65	0.53	0.64	0.52	0.54	0.86	1.10		
Digit span [enkavi2019large]	Memory	0.60	0.59	0.58	0.57	0.57	0.60	0.59	0.58	0.71	0.65	0.62	0.59	0.62	0.59	0.66	0.57	0.55	0.94	2.30		
Episodic long-term memory [popov2023intent]	Memory	1.27	1.05	0.96	0.94	0.90	1.28	1.02	0.92	1.88	1.56	1.37	1.01	1.23	1.05	1.13	0.89	0.87	—	0.69		
N-back [enkavi2019large]	Memory	0.42	0.42	0.41	0.41	0.40	0.42	0.42	0.41	0.49	0.47	0.44	0.41	0.44	0.42	0.52	0.40	0.40	0.58	0.69		
Recall and recognition [cox2018information]	Memory	1.24	1.18	1.13	1.12	1.09	1.22	1.14	1.10	1.69	1.50	1.25	1.15	1.41	1.17	1.38	1.07	1.06	—	—		
Recent probes [enkavi2019large]	Memory	0.26	0.26	0.26	0.26	0.26	0.27	0.26	0.26	0.33	0.28	0.27	0.26	0.27	0.26	0.34	0.26	0.26	0.39	0.69		
		0.40-0.60	0.60-0.80	0.80-1.00	1.00-1.20	1.20-1.50	1.50-2.00	2.00														



Bold+underline = best; underline = second-best (C_p excluded; see below). Lower is better. “—” = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns ([binz2025foundation]): L-70B_p = Llama-3.1-70B base, C_r = our reproduced Centaur-70B, Cog_p = domain-specific cognitive baseline. C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon and Two-step tasks are reported as single merged values by [binz2025foundation].

Rounded to 2 d.p.

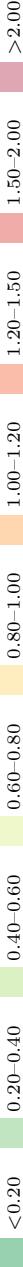
Table 1: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16). Cell colour encodes performance (green = better).

Experiment	Type	Qentaaur				Llama-Centaaur				Smoltaur				Olmotaur				[binz2025foundation]				Chance
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	L-70B _p	C _r	C _p	Cog-p	ln(k)		
Grammar judgement [jansen2021rational]	Misc.	1.47	1.45	1.45	1.44	1.43	1.48	1.45	1.44	1.63	1.54	1.51	1.46	1.51	1.46	1.99	1.44	1.44	1.44	1.41	—	
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.45	2.40	2.36	2.35	2.34	2.56	2.47	2.43	2.67	2.60	2.50	2.45	2.61	2.47	2.64	2.40	2.37	—	—	—	
Serial reaction time task [wu2023chunking]	Misc.	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.16	0.16	0.17	0.16	0.17	0.19	0.18	0.17	0.20	1.39	1.39	
THINGS odd-one-out [hebart2023things]	Misc.	0.82	0.81	0.80	0.79	0.79	0.82	0.82	0.79	1.08	0.90	0.86	0.80	0.86	0.81	1.14	0.80	0.81	0.83	1.10	1.10	
Changing bandit [xiong2023heural]	Bandit	0.30	0.30	0.29	0.29	0.29	0.32	0.31	0.29	0.40	0.38	0.35	0.30	0.39	0.30	0.38	0.45	0.30	0.44	0.69	0.69	
Decisions from experience [wulf2018sampling]	Bandit	0.49	0.49	0.49	0.48	0.50	0.49	0.48	0.54	0.52	0.50	0.49	0.50	0.49	0.49	0.43	0.47	0.37	0.54	—	—	
Drifting four-armed bandit [bahrami2020four]	Bandit	0.73	0.72	0.71	0.71	0.71	0.73	0.72	0.71	0.83	0.78	0.74	0.72	0.74	0.72	0.88	0.71	0.70	0.90	1.39	1.39	
Horizon task [feng2021dynamics]	Bandit	0.23	0.23	0.22	0.22	0.22	0.26	0.24	0.22	0.40	0.34	0.30	0.22	0.35	0.22	0.52	0.40	0.40	0.36	0.69	0.69	
Horizon task [sadeghiyeh2020temporal]	Bandit	0.59	0.58	0.58	0.58	0.58	0.59	0.58	0.58	0.62	0.62	0.60	0.58	0.61	0.58	0.52	0.58	0.40	0.36	0.69	0.69	
Horizon task [somerville2017charting]	Bandit	0.34	0.34	0.34	0.34	0.33	0.35	0.34	0.34	0.48	0.40	0.38	0.34	0.40	0.34	0.52	0.35	0.40	0.36	0.69	0.69	
Horizon task [waltz2020differential]	Bandit	0.15	0.15	0.15	0.15	0.15	0.16	0.16	0.14	0.26	0.21	0.18	0.15	0.19	0.15	0.52	0.15	0.40	0.36	0.69	0.69	
Horizon task [wilson2014humans]	Bandit	0.46	0.46	0.45	0.45	0.45	0.47	0.46	0.45	0.54	0.51	0.49	0.45	0.52	0.45	0.52	0.48	0.40	0.36	0.69	0.69	
Iowa gambling task [steingrover2015data]	Bandit	0.93	0.93	0.91	0.91	0.91	0.94	0.92	0.91	1.01	0.98	0.94	0.92	0.95	0.92	0.99	0.91	0.89	1.16	1.39	1.39	
Prob. instrumental learning [lefevre2017behavioural]	Bandit	0.50	0.50	0.49	0.49	0.49	0.51	0.50	0.49	0.54	0.53	0.51	0.50	0.51	0.50	0.54	0.50	0.49	0.50	0.69	0.69	
Spatially correlated MAB [wu2018generalization]	Bandit	2.01	1.96	1.93	1.87	1.86	2.16	2.00	1.93	2.50	2.30	2.16	1.95	2.26	2.06	2.45	1.82	1.83	2.76	3.40	3.40	
Structured bandit [schulz2020finding]	Bandit	0.67	0.66	0.65	0.65	0.64	0.67	0.66	0.65	0.77	0.71	0.68	0.65	0.73	0.66	0.81	0.64	0.64	1.05	2.08	2.08	
Two-armed bandit [gershman2018deconstructing]	Bandit	0.30	0.30	0.29	0.29	0.29	0.30	0.29	0.29	0.37	0.33	0.31	0.29	0.31	0.30	0.38	0.30	0.30	0.42	0.69	0.69	
Aversive learning [wise2019computational]	Sup. learn.	4.71	4.55	4.27	4.24	3.81	4.58	4.80	3.66	5.35	5.12	4.92	3.93	5.44	4.67	5.11	4.13	4.07	—	—	—	
Medin categorization [levering2020revisiting]	Sup. learn.	0.51	0.51	0.50	0.50	0.50	0.52	0.50	0.50	0.56	0.54	0.52	0.50	0.52	0.51	0.58	0.50	0.50	0.53	—	—	
Multiple-cue judgment [collisio2023numerical]	Sup. learn.	1.17	1.16	1.14	1.14	1.13	1.17	1.15	1.12	1.59	1.49	1.40	1.14	1.67	1.15	1.28	1.14	1.12	1.92	2.20	2.20	
Shepard categorization [badham2017deficits]	Sup. learn.	0.55	0.55	0.54	0.53	0.53	0.56	0.54	0.53	0.62	0.58	0.56	0.54	0.56	0.54	0.58	0.54	0.54	0.61	0.69	0.69	
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.55	0.56	0.54	0.55	0.55	0.55	0.54	0.56	0.58	0.56	0.56	0.55	0.57	0.53	0.57	0.56	0.55	0.63	0.69	0.69	
Mean		0.69	0.68	0.66	0.66	0.64	0.70	0.68	0.64	0.89	0.82	0.73	0.66	0.77	0.68	0.88	0.71	0.69	0.69	0.98	0.98	

Bold + underline = best; underline = second-best (*C_p* excluded; see below). Lower is better. “—” = not available. All models finetuned at LoRA rank 16, bfl6 inference. Reference columns ([binz2025foundation]): L-70B_p = Llama-3.1-70B base, *C_r* = our reproduced Centaur-70B, *C_p* = published Centaur-70B, Cog-*p* = domain-specific cognitive baseline. *C_p* is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. †Horizon and Two-step tasks are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 1: Per-experiment NLL on Psych-101 for all four finetuned families (bfl6, LoRA rank 16), continued. Cell colour encodes performance (green = better).

Experiment	Type	Qwen3 family						Llama-3.1/3.2 family						[binz2025foundation]						Chance				
		0.6B		1.7B		4B		8B		14B		1B		3B		8B		L-70Bp			C _p	C _g p	ln(k)	
		base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft					
Balloon analog risk task [frey2017risk]	Decision	0.09	0.07	0.09	0.07	0.08	0.07	0.08	0.06	0.08	0.06	0.10	0.07	0.08	0.07	0.08	0.06	0.08	0.07	0.06	0.09	0.69		
CPC18 [plonsky2018when]	Decision	0.41	0.35	0.40	0.34	0.40	0.34	0.40	0.34	0.40	0.33	0.43	0.36	0.41	0.35	0.39	0.34	0.41	0.35	0.34	0.66	0.69		
Columbia card task [frey2017risk]	Decision	0.27	0.20	0.26	0.20	0.26	0.20	0.25	0.20	0.25	0.20	0.30	0.21	0.27	0.20	0.25	0.20	0.23	0.21	0.19	0.26	0.69		
Decisions from description [wulff2018sampling]	Decision	1.08	0.62	1.05	0.63	1.01	0.62	0.99	0.61	1.01	0.60	1.55	0.62	1.20	0.63	0.96	0.60	0.76	0.59	0.53	0.61	0.69		
Exponential-symbolic task [garcia2023experiential]	Decision	0.76	0.48	0.68	0.47	0.67	0.46	0.64	0.46	0.57	0.46	0.78	0.47	0.69	0.47	0.64	0.46	0.70	0.46	0.45	—	—		
Gardening task [flesch2018comparing]	Decision	0.52	0.39	0.52	0.39	0.49	0.38	0.47	0.38	0.45	0.38	0.81	0.40	0.64	0.39	0.48	0.38	0.50	0.49	0.38	0.91	0.69		
Go/no-go [enkavi2019large]	Decision	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00		
Inter-temporal choice [ruggieri2022globalizability]	Decision	0.77	0.44	0.80	0.44	0.71	0.44	0.69	0.44	0.70	0.44	0.85	0.44	0.75	0.44	0.76	0.44	0.73	0.44	0.43	0.66	0.69		
Multi-attribute DM [hilbig2014generalized]	Decision	0.49	0.09	0.34	0.08	0.20	0.09	0.19	0.08	0.19	0.08	0.63	0.08	0.45	0.08	0.23	0.08	0.15	0.06	0.06	0.19	0.69		
Risky choice [krueger2024identifying]	Decision	0.92	0.48	0.83	0.47	0.79	0.44	0.74	0.43	0.67	0.42	0.99	0.49	0.85	0.46	0.76	0.43	0.65	0.43	0.43	—	—		
choices13k [petersen2021using]	Decision	0.62	0.44	0.60	0.44	0.59	0.43	0.59	0.43	0.55	0.43	0.67	0.44	0.59	0.44	0.55	0.43	0.53	0.43	0.43	0.66	0.69		
Multi-task RL [tomov2021multitask]	MDP	0.70	0.60	0.72	0.59	0.68	0.58	0.70	0.58	0.68	0.57	0.80	0.60	0.70	0.59	0.66	0.57	0.66	0.57	0.57	1.04	1.10		
Tile-revealing task [kumar2023disentangling]	MDP	2.95	0.02	2.81	0.01	2.67	0.00	2.73	0.07	2.60	0.00	4.86	0.02	3.98	0.00	2.92	0.00	2.74	2.29	1.87	—	—		
Two-step task [kool2017cost]	MDP	0.67	0.54	0.64	0.53	0.64	0.53	0.65	0.53	0.63	0.53	0.67	0.55	0.62	0.53	0.61	0.53	0.61	0.53	0.50	0.60	0.69		
Two-step task [kool2016when]	MDP	0.69	0.51	0.64	0.50	0.64	0.49	0.65	0.49	0.63	0.48	0.67	0.52	0.62	0.50	0.61	0.49	0.61	0.48	0.50	0.60	0.69		
Two-step task [zorowitz2023data]	MDP	0.62	0.52	0.59	0.51	0.60	0.51	0.61	0.51	0.59	0.51	0.64	0.52	0.60	0.51	0.58	0.51	0.61	0.51	0.50	0.60	0.69		
Virtual subway network [tomov2020discovery]	MDP	1.68	1.16	1.55	1.15	1.57	1.14	1.55	1.13	1.57	1.12	1.68	1.19	1.60	1.16	1.53	1.13	1.53	1.16	1.13	—	1.61		
Zoopermarket [ludwig2023human]	MDP	0.61	0.54	0.61	0.52	0.60	0.51	0.60	0.50	0.59	0.48	0.64	0.54	0.62	0.52	0.60	0.50	0.60	0.50	0.48	0.60	0.69		
Cond. assoc. learning [collins2014working]	Memory	0.76	0.54	0.71	0.53	0.71	0.52	0.69	0.52	0.61	0.51	0.82	0.55	0.73	0.53	0.70	0.52	0.64	0.52	0.54	0.86	1.10		
Digit span [enkavi2019large]	Memory	0.94	0.60	0.89	0.59	0.72	0.58	0.70	0.57	0.69	0.57	1.01	0.60	0.84	0.59	0.75	0.58	0.66	0.57	0.55	0.94	2.30		
Episodic long-term memory [popov2023intent]	Memory	1.72	1.27	1.45	1.05	1.33	0.96	1.30	0.94	1.18	0.90	1.97	1.28	1.65	1.02	1.43	0.92	1.13	0.89	0.87	—	0.69		
N-back [enkavi2019large]	Memory	0.58	0.42	0.58	0.42	0.55	0.41	0.55	0.41	0.53	0.40	0.59	0.42	0.57	0.42	0.57	0.41	0.52	0.40	0.40	0.58	0.69		
Recall and recognition [cox2018information]	Memory	1.73	1.24	1.57	1.18	1.44	1.13	1.41	1.12	1.36	1.09	1.87	1.22	1.56	1.14	1.37	1.10	1.38	1.07	1.06	—	—		
Recent probes [enkavi2019large]	Memory	0.70	0.26	0.61	0.26	0.39	0.26	0.41	0.26	0.35	0.26	0.74	0.27	0.56	0.26	0.34	0.26	0.34	0.26	0.26	0.39	0.69		
<0.20	0.20-0.40	0.40-0.60	0.60-0.80	0.80-1.00	1.00-1.20	1.20-1.50	1.50-2.00	>2.00																



Base+underline = best; underline = second-best (C_p excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. “—” = no cognitive baseline reported. Size labels in billions of parameters. Subscript r denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; p denotes values published by [binz2025foundation]. C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 2: Base vs. finetuned NLL on Psych-101 under bf16 inference. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. Cell colour encodes performance; the consistent colour shift from base to ft demonstrates the effect of cognitive finetuning across all scales.

Experiment	Type	Qwen3 family										Llama-3.1/3.2 family										[binz2025foundation]					Chance
		0.6B		1.7B		4B		8B		14B		1B		3B		8B		L-70Bp		C _r	C _p	Cog-p					
		base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft										
Grammar judgement [jansen2021rational]	Misc.	2.12	1.47	2.02	1.45	2.00	1.45	1.89	1.44	1.91	1.43	2.33	1.48	2.16	1.45	2.12	1.44	1.99	1.44	1.44	1.44	1.41	—				
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.75	2.45	2.69	2.40	2.61	2.36	2.57	2.35	2.58	2.34	2.89	2.56	2.80	2.47	2.71	2.43	2.64	2.40	2.37	—	—					
Serial reaction time task [wu2023chunking]	Misc.	0.21	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.31	0.17	0.23	0.17	0.20	0.17	0.19	0.18	0.17	0.20	1.33					
THINGS odd-one-out [hebart2023things]	Misc.	1.21	0.82	1.21	0.81	1.17	0.80	1.15	0.79	1.13	0.79	1.31	0.82	1.18	0.82	1.16	0.79	1.14	0.80	0.81	0.83	1.10					
Changing bandit [xiong2023neural]	Bandit	0.42	0.30	0.38	0.30	0.36	0.29	0.35	0.29	0.34	0.29	1.34	0.32	0.90	0.31	0.39	0.29	0.38	0.45	0.30	0.44	0.69					
Decisions from experience [wulff2021sampling]	Bandit	0.74	0.49	0.66	0.49	0.69	0.49	0.78	0.49	0.61	0.48	0.81	0.50	0.71	0.49	0.65	0.48	0.43	0.47	0.37	0.54	—					
Drifting four-armed bandit [bahrami2020four]	Bandit	0.98	0.73	0.95	0.72	0.92	0.71	0.91	0.71	0.91	0.71	0.95	0.73	0.94	0.72	0.90	0.71	0.88	0.71	0.70	0.90	1.39					
Horizon task [feng2021dynamics]	Bandit	0.47	0.23	0.43	0.23	0.39	0.22	0.36	0.22	0.34	0.22	0.81	0.26	0.67	0.24	0.44	0.22	0.52	0.40	0.40†	0.36†	0.69					
Horizon task [sadeghi2020temporal]	Bandit	0.67	0.59	0.72	0.58	0.66	0.58	0.65	0.58	0.65	0.58	0.67	0.59	0.68	0.58	0.66	0.58	0.52	0.58	0.40†	0.36†	0.69					
Horizon task [somerville2017charting]	Bandit	0.54	0.34	0.55	0.34	0.53	0.34	0.47	0.34	0.49	0.33	0.59	0.35	0.57	0.34	0.51	0.34	0.52	0.35	0.40†	0.36†	0.69					
Horizon task [waltz2020differential]	Bandit	0.33	0.15	0.33	0.15	0.31	0.15	0.27	0.15	0.24	0.15	0.37	0.16	0.37	0.16	0.30	0.14	0.52	0.15	0.40†	0.36†	0.69					
Horizon task [wilson2014humans]	Bandit	0.57	0.46	0.58	0.46	0.56	0.45	0.53	0.45	0.54	0.45	0.64	0.47	0.62	0.46	0.55	0.45	0.52	0.48	0.40†	0.36†	0.69					
Iowa gambling task [steingrover2015data]	Bandit	1.07	0.93	1.06	0.93	1.07	0.91	1.07	0.91	1.07	0.91	1.10	0.94	1.06	0.92	1.04	0.91	0.99	0.91	0.89	1.16	1.39					
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.62	0.50	0.60	0.50	0.57	0.49	0.59	0.49	0.56	0.49	0.62	0.51	0.58	0.50	0.56	0.49	0.54	0.50	0.49	0.50	0.69					
Spatially correlated MAB [wu2018generalization]	Bandit	2.61	2.01	2.55	1.96	2.50	1.93	2.44	1.87	2.46	1.86	2.76	2.16	2.64	2.00	2.58	1.93	2.45	1.82	1.83	2.76	3.40					
Structured bandit [schulz2020finding]	Bandit	0.92	0.67	0.92	0.66	0.88	0.65	0.87	0.65	0.83	0.64	0.94	0.67	0.89	0.66	0.85	0.65	0.81	0.64	0.64	1.05	2.08					
Two-armed bandit [gershman2018deconstructing]	Bandit	0.43	0.30	0.41	0.30	0.40	0.29	0.39	0.29	0.39	0.29	0.43	0.30	0.43	0.29	0.40	0.29	0.38	0.30	0.30	0.42	0.69					
Aversive learning [wise2019computational]	Sup. learn.	5.20	4.71	5.06	4.55	4.95	4.27	4.89	4.24	4.75	3.81	5.66	4.58	5.52	4.80	5.25	3.66	5.11	4.13	4.07	—	—					
Medin categorization [levering2020revisiting]	Sup. learn.	0.60	0.51	0.60	0.51	0.60	0.50	0.59	0.50	0.55	0.50	0.71	0.52	0.60	0.50	0.57	0.50	0.58	0.50	0.50	0.53	—					
Multiple-cue judgment [collsio2023numerical]	Sup. learn.	1.46	1.17	1.38	1.16	1.28	1.14	1.30	1.14	1.26	1.13	1.69	1.17	1.50	1.15	1.32	1.12	1.28	1.14	1.12	1.92	2.20					
Shepard categorization [badham2017deficits]	Sup. learn.	0.63	0.55	0.61	0.55	0.59	0.54	0.58	0.53	0.59	0.53	0.67	0.56	0.62	0.54	0.60	0.53	0.58	0.54	0.54	0.61	0.69					
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.62	0.55	0.59	0.56	0.61	0.54	0.58	0.55	0.58	0.55	0.62	0.55	0.59	0.54	0.58	0.56	0.57	0.56	0.55	0.63	0.69					
Mean (all 48)		0.99	0.69	0.94	0.68	0.90	0.66	0.89	0.66	0.87	0.64	1.14	0.70	1.02	0.68	0.92	0.64	0.88	0.71	0.69	0.69	0.98					

Bold+underline = best; underline = second-best (*C_p* excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better.

"—" = no cognitive baseline reported. Size labels in billions of parameters. Subscript *r* denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; *p* denotes values published by [binz2025foundation]. *C_p* is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. †Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Table 2: Base vs. finetuned NLL on Psych-101 under bf16 inference, continued. Within each model size, the left column (base) shows the pretrained model and the right column (ft) the cognitively finetuned variant.

Task type	Qwentaaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (11)	0.32	0.32	0.32	0.31	<u>0.31</u>	0.32	0.32	<u>0.31</u>	0.40	0.37	0.35	0.32	0.36	0.32	0.46	0.44	0.46	0.30	0.32	0.46
MDP (7)	0.56	0.55	0.54	0.54	0.53	0.56	0.54	<u>0.53</u>	1.03	0.91	0.59	0.54	0.63	0.56	1.07	1.04	1.07	0.79	0.86	0.69
Bandit (13)	0.59	0.59	0.58	<u>0.57</u>	<u>0.57</u>	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (6)	0.72	0.67	0.65	0.64	<u>0.62</u>	0.72	0.66	0.63	0.97	0.85	0.76	0.66	0.77	0.67	0.84	0.78	0.86	0.61	<u>0.62</u>	0.69
Misc. (4)	1.23	1.21	1.19	1.19	<u>1.18</u>	1.26	1.23	1.21	1.39	1.30	1.25	1.22	1.28	1.23	1.45	1.45	1.55	1.20	1.20	0.81
Sup. learn. (5)	1.50	1.47	1.40	1.39	1.30	1.48	1.51	<u>1.27</u>	1.74	1.66	1.59	1.33	1.75	1.48	1.59	1.55	1.66	1.36	1.37	0.92
Mean (46)	0.69	0.68	0.66	0.66	<u>0.64</u>	0.70	0.68	<u>0.64</u>	0.89	0.82	0.73	0.66	0.77	0.68	0.89	0.87	0.92	0.69	0.71	0.69
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																				

Table 3: Mean NLL by task type for finetuned models (bf16). Finetuned families: Qwentaaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Task type	Qwentaaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (9)	0.29	0.29	0.29	0.28	<u>0.28</u>	0.29	0.29	<u>0.28</u>	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	0.27	0.29	0.46
MDP (5)	0.54	0.53	0.52	0.52	<u>0.51</u>	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	0.51	<u>0.52</u>	0.69
Bandit (13)	0.59	0.59	0.58	<u>0.57</u>	<u>0.57</u>	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (4)	0.45	0.45	0.44	0.44	<u>0.44</u>	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	0.44	<u>0.44</u>	0.69
Misc. (3)	0.82	0.81	0.80	0.80	<u>0.80</u>	0.83	0.81	<u>0.80</u>	0.96	0.87	0.84	0.81	0.84	0.81	1.08	1.08	1.16	0.80	0.81	0.81
Sup. learn. (4)	0.69	0.70	0.68	0.68	<u>0.68</u>	0.70	0.68	<u>0.68</u>	0.84	0.79	0.76	0.68	0.83	0.68	0.77	0.75	0.77	0.68	0.68	0.92
Mean (38)	0.53	0.52	0.52	<u>0.51</u>	<u>0.51</u>	0.54	0.52	0.51	0.64	0.59	0.56	0.52	0.59	0.52	0.66	0.65	0.67	0.51	0.52	0.69
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																				

Table 4: Mean NLL by task type for finetuned models (bf16), restricted to the 38 experiments (of 46) for which a cognitive model baseline is available. 8 experiments without a reported cognitive baseline are excluded to enable direct comparison across all columns. Finetuned families: Qwentaaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Task type	Qwentaaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]			Chance
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p	ln(<i>k</i>)
Decision (9)	0.29	0.29	0.29	0.28	<u>0.28</u>	0.29	0.29	<u>0.28</u>	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	0.27	0.29	0.46	0.62
MDP (5)	0.54	0.53	0.52	0.52	<u>0.51</u>	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	0.51	<u>0.52</u>	0.69	0.77
Bandit (12)	0.60	0.60	0.59	<u>0.58</u>	<u>0.58</u>	0.62	0.60	0.58	0.73	0.68	0.64	0.59	0.66	0.60	0.74	0.74	0.76	0.60	0.61	0.75	1.15
Memory (4)	0.45	0.45	0.44	0.44	<u>0.44</u>	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	0.44	<u>0.44</u>	0.69	1.20
Misc. (2)	0.50	0.49	0.48	0.48	<u>0.48</u>	0.50	0.49	<u>0.48</u>	0.62	0.53	0.51	0.48	0.51	0.49	0.67	0.66	0.68	0.49	0.49	0.51	1.24
Sup. learn. (3)	0.75	0.76	0.74	0.74	<u>0.74</u>	0.76	0.74	<u>0.74</u>	0.93	0.87	0.84	0.74	0.93	0.74	0.82	0.81	0.83	0.74	0.74	1.05	1.19
Mean (35)	0.50	0.50	0.49	<u>0.49</u>	<u>0.48</u>	0.51	0.50	0.49	0.61	0.57	0.54	0.49	0.56	0.50	0.63	0.61	0.63	0.49	0.50	0.67	0.97
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																					

Table 5: Mean NLL by task type for finetuned models (bf16), restricted to the 35 experiments (of 46) that have both a reported cognitive model baseline and a well-defined discrete response space admitting a chance-level baseline $\ln(k)$, where k is the number of response options. 11 experiments lacking either a cognitive baseline or a computable $\ln(k)$ (continuous, mixed, or variable action spaces) are excluded. Finetuned families: Qwentaaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (8)	0.53	0.53	0.54	0.54	<u>0.54</u>	0.53	0.53	<u>0.54</u>	0.41	0.46	0.48	0.54	0.47	0.53	0.34	0.35	0.33	0.56	0.52	0.27
MDP (5)	0.28	0.30	0.31	0.31	0.32	0.28	0.30	0.31	0.18	0.20	0.26	0.30	0.20	0.30	0.15	0.17	0.19	0.32	<u>0.31</u>	0.11
Bandit (12)	0.48	0.48	0.49	<u>0.49</u>	<u>0.50</u>	0.46	0.48	0.49	0.36	0.41	0.44	0.49	0.42	0.48	0.35	0.36	0.33	0.46	0.45	0.39
Memory (4)	0.57	0.57	0.58	0.58	<u>0.58</u>	0.56	0.57	0.58	0.46	0.51	0.55	0.57	0.53	0.57	0.42	0.47	0.43	0.58	<u>0.58</u>	0.36
Misc. (2)	0.57	0.57	0.58	0.58	<u>0.58</u>	0.56	0.57	<u>0.58</u>	0.45	0.53	0.55	0.58	0.55	0.57	0.41	0.42	0.40	0.57	0.57	0.55
Sup. learn. (3)	0.30	0.29	0.31	0.31	<u>0.31</u>	0.29	0.31	0.31	0.18	0.23	0.25	0.30	0.20	<u>0.31</u>	0.24	0.24	0.23	0.30	0.30	0.11
Mean (34)	0.46	0.46	0.47	0.48	<u>0.48</u>	0.45	0.47	<u>0.48</u>	0.35	0.39	0.43	0.47	0.40	0.47	0.32	0.34	0.32	0.47	0.46	0.30
<0.05 0.05–0.12 0.12–0.18 0.18–0.25 0.25–0.32 0.32–0.40 0.40–0.48 0.48–0.55 ≥0.55																				

Table 6: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type for finetuned models (bf16). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 34 experiments (of 46) with both a cognitive model baseline and a well-defined discrete response space ($\ln k > 0$); 12 experiments are excluded. Finetuned families: Qwentaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. **Bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Experiment	Type	Qwen 2.5/3 family										Llama 3/3.1/3.2 family										[binz2025foundation]				Chance			
		Base					Non-cog.					Other cog.					Non-cog.												
		8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B	8B	14B		8B	14B	8B
Balloon analog task [frey2017risk]	Decision	0.08	0.08	0.06	0.06	5.87	0.08	0.08	0.08	0.07	0.06	0.10	13.52	0.09	0.17	0.09	0.18	0.07	0.09	0.18	0.07	0.09	0.18	0.07	0.09	0.18	0.07	0.09	0.69
CPC18 [plonsky2018when]	Decision	0.40	0.40	0.34	0.33	8.54	0.40	0.41	0.39	0.35	0.34	0.47	14.34	0.44	0.95	0.42	0.64	0.35	0.44	0.95	0.42	0.64	0.35	0.44	0.95	0.42	0.64	0.69	
Columbia card task [frey2017risk]	Decision	0.25	0.25	0.20	0.20	6.72	0.26	0.27	0.25	0.20	0.20	0.26	14.07	0.27	0.48	0.26	0.49	0.21	0.27	0.48	0.26	0.49	0.21	0.27	0.48	0.26	0.49	0.69	
Decisions from description [wulff2018sampling]	Decision	0.99	1.01	0.61	0.60	10.13	1.02	1.20	0.96	0.63	0.60	1.28	13.59	1.34	2.99	1.17	4.44	0.59	1.34	2.99	1.17	4.44	0.59	1.34	2.99	1.17	4.44	0.61	
Experiential-symbolic task [garcia2023experiential]	Decision	0.64	0.57	0.46	0.46	7.86	0.58	0.69	0.64	0.47	0.46	0.73	12.95	0.79	1.37	0.67	0.95	0.46	0.79	1.37	0.67	0.95	0.46	0.79	1.37	0.67	0.95	—	
Gardening task [flesch2018comparing]	Decision	0.47	0.45	0.38	0.38	10.28	0.45	0.64	0.48	0.39	0.38	0.48	16.26	0.53	0.61	0.46	0.64	0.49	0.53	0.61	0.46	0.64	0.49	0.53	0.61	0.46	0.64	0.91	
Go/no-go [enkavi2019large]	Decision	0.01	0.00	0.00	0.00	4.24	0.00	0.01	0.00	0.00	0.00	0.01	12.55	0.01	0.04	0.00	0.01	0.00	0.01	0.04	0.00	0.01	0.00	0.01	0.04	0.00	0.01	0.00	
Intertemporal choice [ruger2022globalizability]	Decision	0.69	0.70	0.44	0.44	12.23	0.74	0.75	0.76	0.44	0.44	0.76	13.82	0.72	2.75	1.04	2.72	0.44	0.72	2.75	1.04	2.72	0.44	0.72	2.75	1.04	2.72	0.66	
Multi-attribute DM [hibig2014generalized]	Decision	0.19	0.19	0.08	0.08	10.45	0.20	0.45	0.23	0.08	0.08	0.30	13.86	0.45	0.61	0.23	0.54	0.06	0.45	0.61	0.23	0.54	0.06	0.45	0.61	0.23	0.54	0.19	
Risky choice [krueger2024identifying]	Decision	0.74	0.67	0.43	0.42	7.17	0.67	0.85	0.76	0.46	0.43	0.85	16.10	0.94	1.79	0.82	1.35	0.43	0.94	1.79	0.82	1.35	0.43	0.94	1.79	0.82	1.35	—	
choices13k [petersen2021using]	Decision	0.59	0.55	0.43	0.43	9.40	0.53	0.59	0.55	0.44	0.43	0.68	13.44	0.65	1.65	0.60	1.13	0.43	0.65	1.65	0.60	1.13	0.43	0.65	1.65	0.60	1.13	0.66	
Multi-task RL [tomov2021multitask]	MDP	0.70	0.68	0.58	0.57	7.95	0.66	0.70	0.66	0.59	0.57	0.82	14.70	0.81	1.20	0.70	1.06	0.57	0.81	1.20	0.70	1.06	0.57	0.81	1.20	0.70	1.06	1.10	
Tile-revealing task [kumar2023disentangling]	MDP	2.73	2.60	0.07	0.00	35.15	2.62	3.98	2.92	0.00	0.00	3.03	65.75	2.93	3.48	2.85	3.37	2.29	2.93	3.48	2.85	3.37	2.29	2.93	3.48	2.85	3.37	—	
Two-step task [kool2017cost]	MDP	0.65	0.63	0.53	0.53	7.90	0.63	0.62	0.61	0.53	0.53	0.71	14.06	0.67	1.17	0.67	0.81	0.53	0.67	1.17	0.67	0.81	0.53	0.67	1.17	0.67	0.81	0.69	
Two-step task [kool2016when]	MDP	0.65	0.63	0.49	0.48	8.82	0.63	0.62	0.61	0.50	0.49	0.72	14.63	0.68	1.15	0.67	0.82	0.48	0.68	1.15	0.67	0.82	0.48	0.68	1.15	0.67	0.82	0.69	
Two-step task [zorewicz2023data]	MDP	0.61	0.59	0.51	0.51	7.68	0.58	0.60	0.58	0.51	0.51	0.65	13.18	0.63	1.03	0.62	0.82	0.51	0.63	1.03	0.62	0.82	0.51	0.63	1.03	0.62	0.82	0.69	
Virtual subway network [tomov2020discovery]	MDP	1.55	1.57	1.13	1.12	11.39	1.60	1.60	1.53	1.16	1.13	1.61	18.21	1.69	2.41	1.64	2.10	1.16	1.69	2.41	1.64	2.10	1.16	1.69	2.41	1.64	2.10	1.61	
Zooermarket [ludwig2023human]	MDP	0.60	0.59	0.50	0.48	6.62	0.59	0.62	0.60	0.52	0.50	0.62	13.75	0.65	1.49	0.63	1.09	0.50	0.65	1.49	0.63	1.09	0.50	0.65	1.49	0.63	1.09	0.69	
Cond. assoc. learning [collins2014working]	Memory	0.69	0.61	0.52	0.51	10.04	0.59	0.73	0.70	0.53	0.52	0.72	16.16	0.80	0.92	0.68	0.95	0.52	0.80	0.92	0.68	0.95	0.52	0.80	0.92	0.68	0.95	1.10	
Digit span [enkavi2019large]	Memory	0.70	0.69	0.57	0.57	2.96	0.70	0.84	0.75	0.59	0.58	0.83	4.65	0.87	2.16	0.83	1.02	0.57	0.87	2.16	0.83	1.02	0.57	0.87	2.16	0.83	1.02	0.94	
Episodic long-term memory [popov2023intent]	Memory	1.30	1.18	0.94	0.90	9.08	1.21	1.65	1.43	1.02	0.92	1.52	17.12	1.82	3.07	1.72	2.22	0.89	1.82	3.07	1.72	2.22	0.89	1.82	3.07	1.72	2.22	2.30	
N-back [enkavi2019large]	Memory	0.55	0.53	0.41	0.40	9.24	0.53	0.57	0.57	0.42	0.41	0.57	14.42	0.59	0.79	0.57	0.75	0.40	0.59	0.79	0.57	0.75	0.40	0.59	0.79	0.57	0.75	0.69	
Recall and recognition [cox2018information]	Memory	1.41	1.36	1.12	1.09	9.33	1.38	1.56	1.37	1.14	1.10	1.53	24.31	1.65	4.16	1.61	2.60	1.07	1.65	4.16	1.61	2.60	1.07	1.65	4.16	1.61	2.60	—	
Recent probes [enkavi2019large]	Memory	0.41	0.35	0.26	0.26	9.92	0.34	0.56	0.34	0.26	0.26	0.40	14.16	0.61	1.00	0.39	1.06	0.26	0.61	1.00	0.39	1.06	0.26	0.61	1.00	0.39	1.06	0.69	
0.20-0.40						0.60-0.80						1.00-1.20		1.20-1.50		1.50-2.00		>2.00											

Bold+underline = best; underline = second-best. Lower is better. “—” = not available. Models are grouped by base-model family. “Base” = pretrained model without finetuning (bf16); “Other cog.” = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; “Non-cog.” = models finetuned for general instruction-following or reasoning. **Herm.** = Hermes [teknium2024hermes]; **Hemo.** = Nemotron [nvidia2025nemotron]; **Socr.** = Socrates [kolluri2025finetuning]; **Be.FM** = [xie2025fm]. Base model versions differ within families: **Qwen** uses Qwen3; **Socr.**-14B uses Qwen2.5; **Herm.**-14B uses Qwen2.5. Llama-Centaur uses Llama-3.1/3.2; **Socr.**-8B uses Llama-3; **Be.FM** uses Llama-3.1.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family. Within each family, **Qwen**taur/Llama-Centaur (cognitive SFT on Psych-101) consistently outperform both other behavioural FMs and non-cognitive controls, demonstrating that alignment with human behavioural patterns requires cognitive finetuning on structured experimental data, not merely finetuning per se.

Experiment	Type	Queen 2.5/3 family										Llama 3/3.1/3.2 family										[binz2025foundation]						Chance					
		Base					Non-cog.					Other cog.					Non-cog.					Cog.p											
		8B		14B		Base	8B		14B		Non-cog.	8B		14B		Other cog.	8B		14B		Non-cog.	8B		14B		Cog.p	ln(k)						
		8B	14B	8B	14B		8B	14B	8B	14B		8B	14B	8B	14B		8B	14B	8B	14B		8B	14B	8B	14B		8B		14B	8B	14B	8B	14B
Grammar judgement [jansen2021rational]	Misc.	1.89	1.91	1.44	1.43		10.37	1.94	2.16	2.12	1.45	1.44	2.11	12.88	2.27	3.90	2.53	3.12	1.44	1.41													
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.57	2.58	2.35	2.34		5.09	2.59	2.80	2.71	2.47	2.43	2.84	3.40	2.98	4.05	2.97	3.23	2.40	—													
Serial reaction time task [wu2023chunking]	Misc.	0.19	0.19	0.17	0.17		2.92	0.19	0.23	0.20	0.17	0.17	0.25	14.43	0.21	0.25	0.20	0.26	0.18	—													
THINGS odd-one-out [hebart2023things]	Misc.	1.15	1.13	0.79	0.79		12.15	1.15	1.18	1.16	0.82	0.79	1.19	14.31	1.24	2.13	1.28	1.75	0.80	0.83	1.10												
Changing bandit [xiong2023neural]	Bandit	0.35	0.34	0.29	0.29		7.52	0.34	0.90	0.39	0.31	0.29	0.42	14.76	0.39	0.78	0.36	0.53	0.45	0.44	0.69												
Decisions from experience [wulff2018sampling]	Bandit	0.78	0.61	0.49	0.48		7.18	0.64	0.71	0.65	0.49	0.48	0.83	13.74	0.69	2.29	0.88	1.78	0.47	0.54	—												
Drifting four-armed bandit [bahrami2020four]	Bandit	0.91	0.91	0.71	0.71		8.43	0.90	0.94	0.90	0.72	0.71	0.99	13.91	0.99	2.10	0.96	1.78	0.71	0.90	1.39												
Horizon task [feng2021dynamics]	Bandit	0.36	0.34	0.22	0.22		7.55	0.35	0.67	0.44	0.24	0.22	0.39	15.10	0.43	0.65	0.36	0.62	0.40	0.36	0.69												
Horizon task [sadeghiyeh2020temporal]	Bandit	0.65	0.65	0.58	0.58		8.62	0.66	0.68	0.66	0.58	0.58	0.69	14.94	0.73	1.10	0.69	0.93	0.58	0.36	0.69												
Horizon task [somerville2017charting]	Bandit	0.47	0.49	0.34	0.33		8.49	0.49	0.57	0.51	0.34	0.34	0.53	15.18	0.57	0.86	0.47	0.74	0.35	0.36	0.69												
Horizon task [waltz2020differential]	Bandit	0.27	0.24	0.15	0.15		7.82	0.26	0.37	0.30	0.16	0.14	0.33	15.42	0.35	0.54	0.26	0.54	0.15	0.36	0.69												
Horizon task [wilson2014humans]	Bandit	0.53	0.54	0.45	0.45		8.06	0.54	0.62	0.55	0.46	0.45	0.57	15.39	0.62	0.89	0.56	0.79	0.48	0.36	0.69												
Iowa gambling task [steingrover2015data]	Bandit	1.07	1.07	0.91	0.91		8.45	1.06	1.06	1.04	0.92	0.91	1.11	13.23	1.13	2.27	1.09	1.60	0.91	1.16	1.39												
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.59	0.56	0.49	0.49		8.30	0.53	0.58	0.56	0.50	0.49	0.61	15.92	0.68	1.16	0.54	0.86	0.50	0.50	0.69												
Spatially correlated MAB [wu2018generalization]	Bandit	2.44	2.46	1.87	1.86		3.14	2.48	2.64	2.58	2.00	1.93	2.82	5.43	2.64	3.93	2.78	3.19	1.82	2.76	3.40												
Structured bandit [schulz2020finding]	Bandit	0.87	0.83	0.65	0.64		1.57	0.80	0.89	0.85	0.66	0.65	0.98	2.12	0.91	1.41	0.90	1.08	0.64	1.05	2.08												
Two-armed bandit [gershman2018deconstructing]	Bandit	0.39	0.39	0.29	0.29		7.30	0.41	0.43	0.40	0.29	0.29	0.46	13.99	0.47	0.98	0.44	0.73	0.30	0.42	0.69												
Aversive learning [wise2019computational]	Sup. learn.	4.89	4.75	4.24	3.81		7.19	4.78	5.52	5.25	4.80	3.66	5.49	18.42	5.52	7.03	5.33	6.14	4.13	—	—												
Medin categorization [levering2020revisiting]	Sup. learn.	0.59	0.55	0.50	0.50		11.00	0.56	0.60	0.57	0.50	0.50	0.63	13.53	0.63	1.15	0.62	0.90	0.50	0.53	—												
Multiple-cue judgment [collisio2023numerical]	Sup. learn.	1.30	1.26	1.14	1.13		6.92	1.27	1.50	1.32	1.15	1.12	1.38	18.06	1.51	2.56	1.37	1.85	1.14	1.92	2.20												
Shepard categorization [badham2017deficits]	Sup. learn.	0.58	0.59	0.53	0.53		10.13	0.59	0.62	0.60	0.54	0.53	0.60	12.66	0.64	0.92	0.63	0.76	0.54	0.61	0.69												
Weather prediction task [speckenbrink2008learning]	Sup. learn.	0.58	0.58	0.55	0.55		10.65	0.56	0.59	0.58	0.54	0.56	0.62	17.01	0.63	0.89	0.61	0.77	0.56	0.63	0.69												
Mean (all 46)		0.89	0.87	0.66	0.64		8.65	0.87	1.02	0.92	0.68	0.64	0.99	15.07	1.02	1.72	0.98	1.43	0.71	0.69	0.98												

Bold+underline = best; **underline** = second-best. Lower is better. “—” = not available. Models are grouped by base-model family. “Base” = pretrained model without finetuning (bf16); “Other cog.” = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; “Non-cog.” = models finetuned for general instruction-following or reasoning. *Herm.* = Hermes [tekinium2024hermes]; *Nemo.* = Nemotron [nvidia2025nemotron]; *Socr.* = Socrates [kolluri2025socrates]; *Be.FM* = [xie2025fm]. Base model versions differ within families: *Qwen* uses Qwen3; *Socr.*-14B uses Qwen2.5; *Herm.*-14B uses Qwen2.5. *Llama-Centaur* uses Llama-3.1/3.2; *Socr.*-8B uses Llama-3; *Be.FM* uses Llama-3.1.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family (continued).